United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Mid Exam Spring 2023

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: **30** Duration: 1 hour 45 Minutes

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. (a) Show the corresponding inputs and outputs from **Lexical analyzer** to **Code Optimizer** phase of compiler based on the following expression where all the variables are **int** type.

$$Y = (M * N\text{^}P / Q)\text{^}Z - 2 + 5;$$

1 2 3 4 5 6

(b) CSE-4611 course, which is taught at UIU, is all about compiler design. Compiler is a type of language processors. Draw a block diagram showing all the phases of compiler and explain the working of each phases briefly. (You do not need to give an example for that)

(c) Symbol table is a data structure that is used by all phases of a compiler. Give a brief description of symbol table.

(d) **"Interpreter is better error detector than Compiler"**. Agree or Disagree? Justify.

2. Write down the three address representation of the following expression.

z = x − ((((a^3)^2^1^3)) − (b/3/p^r) − a/2^d)

3. Draw Syntax Tree for the following expression.

$$x = b * (3 \text{ ^ } 5 + 2 * (4 \text{ ^ } 2)) - 5/((z - 1) * 2)/(d - 2 * 3 + 2 \text{ ^ } 1 \text{ ^ } 3)$$

4. Prove that the following expression **(0 ∪ (10) * 1) *** can be represent by the grammar given below.

E → E*| (E) | A

A → A ∪ A | AA | B | D

B → B * B | BB | E | ∈

D → 0 | 1 | ∅ | ∈

5. Convert the following infix expression to a postfix expression.

((x+5)^3+2)^y^z^x/(3/2^2+a)^((z+1)*3^(x+y+z))

6. Convert the following infix expression to a prefix expression.

a*(8^a-5^(z^3))^((z^4-6*a)*3^a)/((w+1)*(a+b+z))

7. Evaluate the following prefix expression.

+^*^2321//^424^+/6^212-*+813^24